Diffusion-Ordered NMR Spectroscopy: A Versatile Tool for the Molecular Weight Determination of Uncharged Polysaccharides

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Diffusion-ordered NMR spectroscopy (DOSY) experiments have been carried out on dilute aqueous solutions of uncharged saccharide systems and, in particular, on six well characterized pullulan fractions of different molecular weights. The values of diffusion coefficients and hydrodynamic radii determined for the pullulan fractions are in good agreement with the results obtained with other methodologies such as light scattering. Fitting the diffusion coefficients data as a function of the molecular weight allows for the determination of a calibration curve that can be applied to a wide range of mono-, oligo-, and polysaccharides. Therefore, DOSY is proposed as a versatile tool for achieving a simple estimation of the molecular weight of uncharged polysaccharides. Mixtures of homopolymers of different molecular weight can be nicely separated. An advantage of the method is that the same sample used for the NMR characterization can be used for the molecular weight determination without any further manipulation. Other water soluble polymers, such as poly(ethylene oxide) and poly(vinylpyrrolidone), can be roughly characterized using the same calibration curve.

Introduction

Polysaccharides together with proteins and lipids constitute a major class of biomacromolecules and play key roles in biological recognition processes[1] and in many fields of pharmaceutical and food industry. Nevertheless, because of difficulties in synthesis and purification, many aspects of carbohydrate chemistry are still poorly known.[2] NMR spectroscopy is one of the most widely used techniques to characterize these macromolecules. Many NMR techniques together with new computational methods have been developed to aid in the structural elucidation of polysaccharides.[3] A full characterization of the investigated polysaccharide most often requires not only the structural assignment but also the determination of the molecular weight, which is of the utmost importance because it is closely related to the industrial properties of the polysaccharide; for instance, the molecular weight together with the molecular weight distribution play a determinant role in the formation of gels. However, common chemical reactions in polysaccharide chemistry such as oxidization, depletion, derivatization and epimerization obviously modify the polysaccharide but can also drastically lower the molecular weight, thereby compromising its physical properties. Several techniques such as photonic correlation spectroscopy (PCS), gel permeation chromatography (GPC), and dynamic light scattering (DLS) are usually employed to determine the molecular weight of polysaccharides. These methods work well on rather small amounts of material, but usually imply specific sample manipulations (filtrations, centrifugations, etc.) which could become severely limiting when working with polysaccharides that are difficult to manipulate. On the other hand, PCS cannot be applied satisfactorily on low molecular weight fractions.[4]

Diffusion-ordered NMR spectroscopy (DOSY)[5] was initially designed for the analysis and characterization of mixtures and aggregates.[6] DOSY has also been used for the study of intermolecular interactions[7,8] and was recently applied to optimize the purification procedure in the synthesis of hyaluronate derivatives.[9] More specifically, DOSY has been shown to allow for the determination of molecular size distribution[10] and applied to the study of phospholipid vesicles.[11] In addition, Chen et al.[12] have demonstrated that DOSY can be applied to the determination of molecular weight distributions for polymers by performing DOSY experiments on poly(ethylene oxide) fractions. Finally, Jerschow et al.[13] have demonstrated its applicability to the study of polymer mixtures and polymer blends.

Here, we propose the DOSY technique as a versatile NMR tool to determine the average molecular weight of uncharged water soluble oligo- and polysaccharides. The diffusion of polymers is well understood and has been extensively studied.[14] In particular, Callaghan et al. have investigated the self-diffusion properties of dextran in water solutions[15] by means of pulsed-field gradient NMR. However, to our knowledge, the idea of using NMR diffusion data achieved with the DOSY technique as a simple and fast way to estimate the average molecular weight of polysaccharides has not been reported so far. To this purpose, different

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fractions of a recognized model for linear polysaccharides, pullulan, were analyzed by DOSY and the data were fitted to obtain a calibration curve whose reliability was then checked by measuring the diffusion coefficients of other uncharged saccharidic systems of known molecular weights.

Experimental Section

Pullulan samples of known molecular weights (5.8, 12.2, 28.3, 100, 180, and 853 kDa) were supplied by Shodex (Showa Denko company, Kawasaki city Kanagawa, Japan). Dextran fractions (M_W = 1.3, 5.2, 148, 273, 410, and 668 kDa) certified according to DIN were purchased from Fluka. Another dextran fraction (M_W = 68 kDa) was purchased from Sigma. The sucrose (M_W = 342 Da), α-, β-, and γ-cyclodextrins (M_W = 973, 1136, and 1297 Da), and amylose (M_W = ca. 21 kDa) samples were supplied by Fluka. Finally, the oligosaccharide (M_W = 3 kDa), a branched mannose derivative, was kindly provided by Dr. Porro from Biosynth (Siena, Italy). All saccharidic systems studied were characterized by a good water solubility and a narrow molecular weight distribution. Other uncharged water soluble polymers, such as poly(ethylene oxide) (M_W = 26, 85, and 885 kDa) and poly(vinylpyrrolidone) (M_W = 10, 24, and 40 kDa), were also studied and purchased from Sigma. All products were used without any further characterization.

Typically, 0.8 mg of sample was dissolved in 700 μL of D_2O. ^1H and ^1H detected DOSY experiments were performed at 300 K on a Bruker AVANCE AQS600 NMR spectrometer operating at 600.13 MHz and equipped with a Bruker multinuclear z-gradient inverse probehead capable of producing gradients in the z direction with a strength of 55 G cm^{-1}.

DOSY experiments were performed using a stimulated-echo sequence incorporating bipolar gradient pulses and a longitudinal eddy current delay (BPP-LED).[16] When necessary, the HOD residual signal was suppressed by means of a soft presaturation. The gradient strength was logarithmically incremented in 32 steps from 2% up to 95% of the maximum gradient strength. Diffusion times and gradient pulse durations were optimized for each experiment in order to achieve a 95% decrease in the resonance intensity at the largest gradient amplitude; typically, diffusion times between 400 and 1000 ms and bipolar rectangular gradient pulses between 1.0 and 2.3 ms were employed. The longitudinal eddy current delay was held constant to 25 ms, whereas the gradient pulse recovery time was set to 100 μs. After Fourier transformation and baseline correction, the diffusion dimension of the 2D DOSY spectra was processed by means of the Bruker Xwinnmr software package (version 3.5).

Results and Discussion

DOSY experiments have been initially performed on dilute aqueous solutions of six pullulan fractions of different molecular weights. Pullulan is a water soluble linear polysaccharide consisting of α-1,6 linked maltotrioses. It has been widely studied[4,17−19] and proposed as a standard sample to investigate the proprieties of linear water soluble polysaccharides. In water, pullulan has been shown to behave as a random coil.[4,17]

A DOSY experiment yields a pseudo 2D spectrum with NMR chemical shifts on one dimension (horizontal axis) and self-diffusion coefficients on the other one (vertical axis). In other words, a diffusion spectrum at each chemical shift can be obtained and the diffusion coefficient of the investigated compound can be measured. DOSY is one of many other available methods used to analyze pulsed field gradient NMR diffusion experiments.[20] It has the advantage over other methods in that it allows for the determination of the diffusion coefficients of all compounds in the mixture and displays the information in a convenient pictorial form.[21]

Commonly available softwares for processing DOSY data allow the use of mono and biexponential fitting routines to process the diffusion dimension. Therefore, a continuous distribution of molecular weights such as that usually found in polymeric systems might be of difficult handling.[12] In other words, DOSY, such as other techniques, might require a previous GPC preparation in order to isolate and collect monodisperse polysaccharidic fractions on which more accurate determinations of M_W can be achieved. For our purpose, samples with relatively narrow molecular weight distribution have been studied.

In Figure 1, the 2D DOSY spectra of three pullulan fractions of different molecular weights are shown.

These bidimensional maps show clearly that the diffusion coefficient of deuterated water remains constant, whereas the diffusion coefficient of the pullulan fractions decreases with the increase in molecular weight. The values of the diffusion coefficients are more precisely obtained by fitting the exponential decay of the resonance signal intensity against the gradient strength. The results are reported in Table 1.

According to the method reported by Nischinari et al.,[4] the double-logarithmic plot of the self-diffusion coefficient D of the polysaccharide in dilute solutions against the weight-average molecular weight M_w is reported in Figure 2a. In such a plot, the M_w values of the pullulan fractions are all located on a straight line for the whole range of molecular weights considered in this study. A least-squared fitting of the data leads to the relationship

$$D = 8.2 \times 10^{-9} M_W{}^{-0.49}\ (\mathrm{m^2\ s^{-1}}) \qquad (1)$$

In eq 1, the exponent value of 0.49 ± 0.01 is in agreement with the Zimm model, which predicts for flexible polymers an exponent value of 0.5 in a θ solvent and 0.6 in a good solvent. The obtained value is relatively different from the 0.55 value usually observed for flexible polymers in good solvent, but agrees satisfactorily well with the 0.51 value reported by Nishinari's group, even though they used two different techniques, namely, hydrodynamic methods and photon correlation spectroscopy, for low and high molecular weights, respectively.[4] With respect to both techniques, DOSY has the advantage to allow for the measurement of the diffusion coefficient over a wider range of molecular weights.

The translational diffusion coefficient of a particle is related to its hydrodynamic radius according to the Stokes–Einstein equation. The diffusion coefficients obtained from the DOSY experiments were used to evaluate the hydrody-

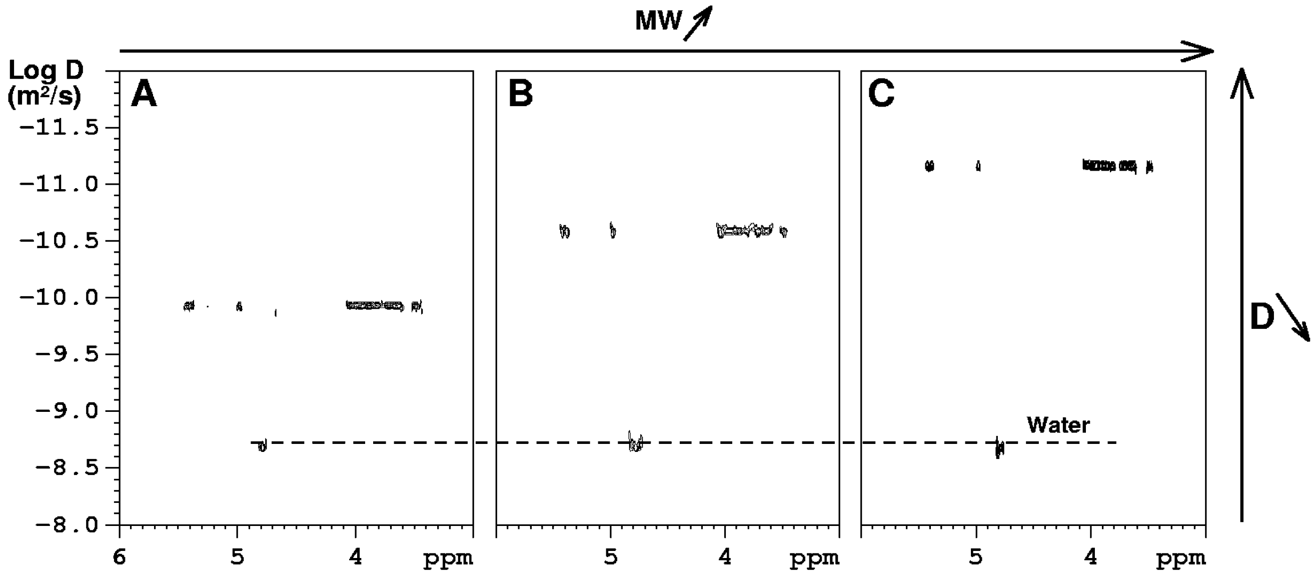

Figure 1. 2D DOSY spectra recorded in D_2O at 300 K of three pullulan fractions with different molecular weights: $M_W = 5.8$ kDa (1a), $M_W = 100$ kDa (1b), and $M_W = 853$ kDa (1c).

Table 1. Diffusion Coefficients and Hydrodynamic Radii (Stokes−Einstein Model) of Six Dilute Aqueous Solutions of Pullulan Determined by DOSY Experiments at 300 K[a]

pullulan fraction	M_W (g mol^{-1})	self-diffusion coefficient D (m^2 s^{-1})		pullulan hydrodynamic radius r_H (nm)		
		D_2O ($\times 10^{-9}$)	pullulan ($\times 10^{-11}$)	r_H[b]	r_H[c]	
1	5800	2.02 ± 0.02	11.6 ± 0.3	1.9	2.0	(5300)
2	12 000	2.02 ± 0.02	8.0 ± 0.4	2.7	2.6	(12 300)
3	28 300	2.02 ± 0.02	5.7 ± 0.3	3.9	4.3	(23 600)
4	100 000	2.01 ± 0.02	2.8 ± 0.2	7.8	8.4	(112 000)
5	180 000	2.02 ± 0.02	2.2 ± 0.2	10.0	11.1	(228 000)
6	853 000	2.01 ± 0.02	1.05 ± 0.1	20.9	25.3	(1 015 000)

[a] Hydrodynamic radii determined with other methods are reported in the last column. [b] As obtained in this work. [c] As obtained on pullulan fractions whose M_W values are indicated between brackets, from ref 4.

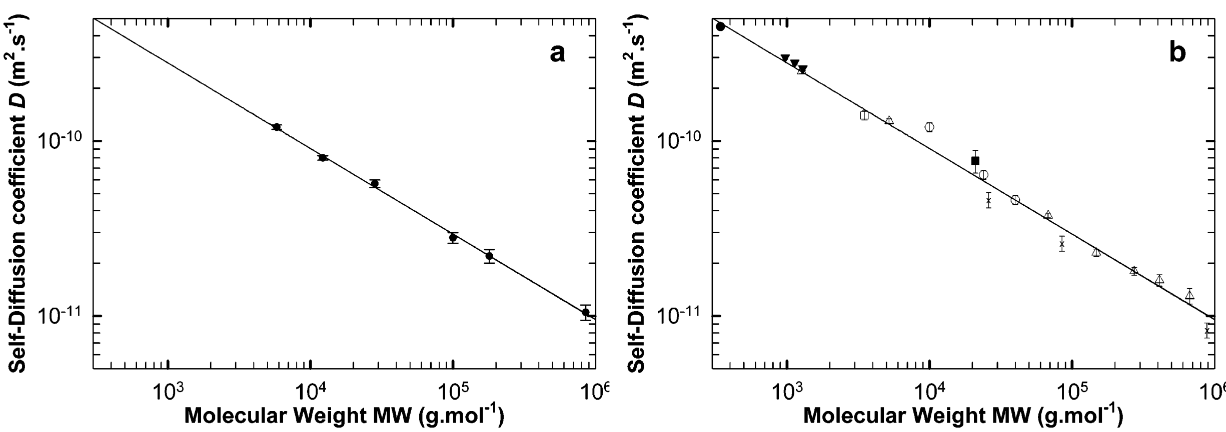

Figure 2. (a) Double-logarithmic plot of D against M_W for six pullulan fractions as determined by DOSY experiments recorded at 300 K in dilute D_2O solution. All data points are located on a straight line. (b) Double-logarithmic plot of D against M_W for some dextran (△), sucrose (●), and cyclodextrin (▼) samples. A branched mannose derivative oligosaccharide (□) and amylose (■) samples, together with some poly(ethylene oxide) (×) and poly(vinylpyrrolidone) (○) samples, are also indicated. The straight line is the calibration curve determined in Figure 2a.

namic radius of the particles for all six pullulan fractions. It should be emphasized at this point that an accurate determination of hydrodynamic radii can only be achieved by extrapolation of diffusion data to infinite dilution as previously reported by Nishinari et al.[4] However, the results reported in Table 1 show a relatively good agreement with those reported by Nishinari et al. for pullulan fractions of similar molecular weights at infinite dilution. This confirms that the solutions used for the DOSY experiments were sufficiently diluted. In addition, for all pullulan fractions, the diffusion coefficient of deuterated water remains equal within experimental error to its value measured at 300 K.

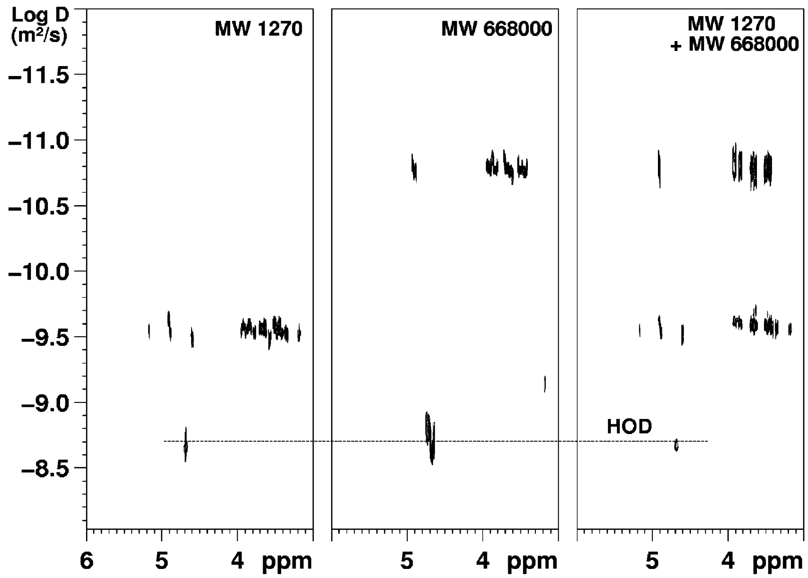

Figure 3. 2D DOSY spectra recorded in D_2O at 300 K of two dextran fractions with M_W = 1270 and 668 000 g mol^{-1}, respectively, and their 1:1 (w/w) mixture.

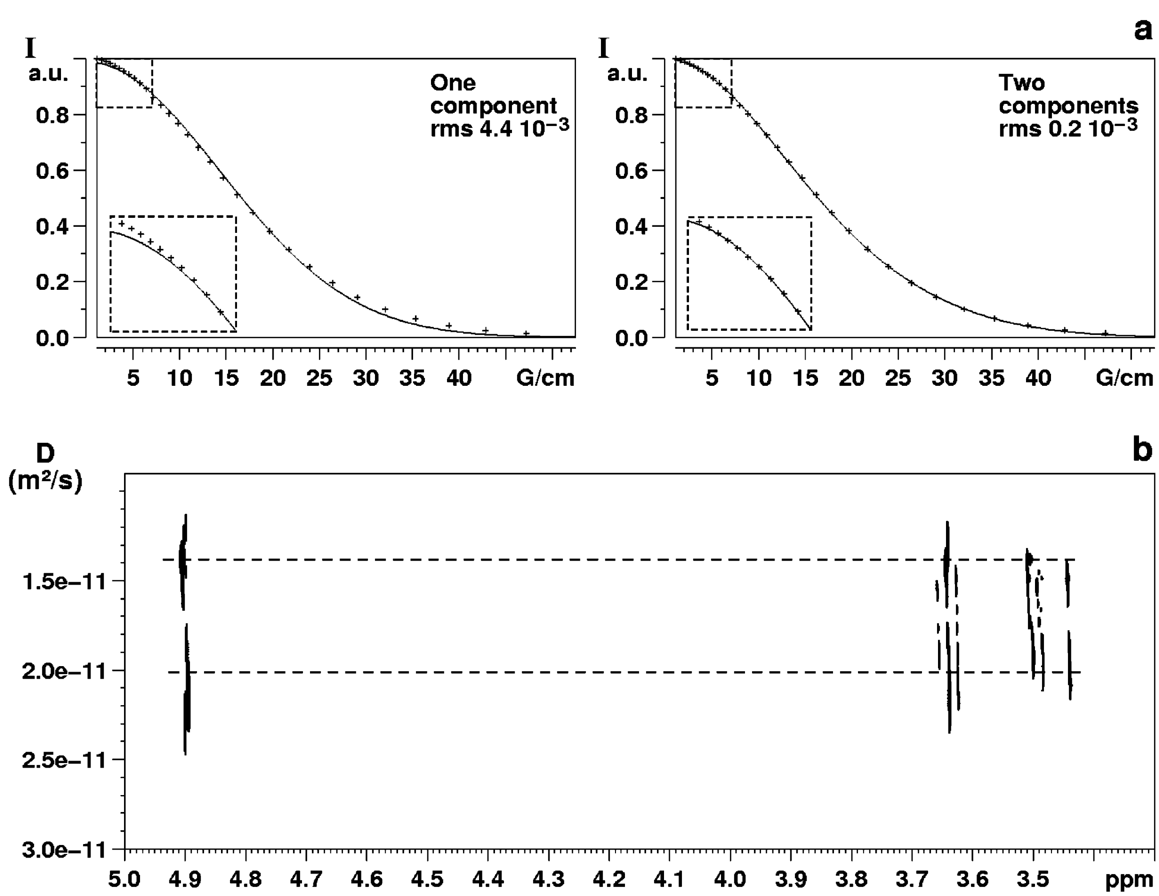

Figure 4. Diffusion data of a 1:1 (w/w) mixture of two dextran fractions (M_W = 270 and 670 kDa) recorded at 300 K in D_2O. (a) Plot of the normalized resonance intensity (δ = 4.90 ppm) as a function of the gradient strength and corresponding exponential fitting with one and two components, respectively. The root mean square (rms) of both fits is reported. (b) DOSY spectrum of the same mixture: two components can be clearly distinguished in the diffusion dimension. Their diffusion coefficient matches well the corresponding value of the separate components.

This means that, as expected, no retarding effects on the solvent diffusion are detectable and hence confirms again that the solutions we have used were sufficiently diluted.

It should be pointed out that the Zimm parameters and, in turn, the corresponding calibration curve derived in Figure 2a depend directly on the type of polysaccharide under investigation, i.e., pullulan in this case. However, we may ask whether such a calibration curve could be applied to evaluate the molecular weight of other saccharidic systems.

To address this issue, dilute solutions of mono-, oligo-, and polysaccharidic samples have been analyzed by DOSY, and the results have been compared with the calibration curve derived from the study of pullulan. In particular, DOSY experiments have been carried out on dextran fractions of known molecular weights. Dextran consists mainly of α-1,6 linked D-glucose with α-1,3 branchings. In addition, a set of low and intermediate molecular weight compounds have been analyzed: sucrose; α-, *β*-, and *γ*-cyclodextrins; and a mannose-rich oligosaccharide. The measured diffusion coefficients with the corresponding molecular weights of all analyzed compounds are reported in Figure 2b together with the calibration curve determined in Figure 2a. The agreement between the calibration curve and the experimental data points is fairly good except maybe for the amylose sample, for which the uncertainty on the molecular weight is intrinsically high. Again the diffusion coefficient values are in agreement with previously reported data.[15,22] Therefore, diffusion data such as that provided by DOSY can be used to provide a rough estimation of the molecular weights for linear and slightly branched water soluble uncharged polysaccharides over a really wide range of molecular weights.

Figure 2b includes also the diffusion coefficients of two other water soluble polymers, namely, poly(ethylene oxide) and poly(vinylpyrrolidone). As far as PEO is concerned, the agreement is rather poor; in fact, in the double-logarithmic plot, all three samples seem aligned on a straight line parallel to the calibration curve. Two poly(vinylpyrrolidone) samples show a good agreement, whereas one is clearly out of the calibration curve. As a conclusion, we may affirm that the calibration curve can be applied to a large class of uncharged water soluble polymers even though the error is likely to be higher when it is applied to non-saccharidic systems.

A further advantage of DOSY is the possibility of resolving mixtures of homopolymers with different molecular weights.

Figure 3 illustrates how a mixture of two dextran fractions of different molecular weights (M_w 1.27 and 668 kDa) can be resolved in the diffusion dimension, allowing the discrimination of their molecular weights through the use of the previously determined calibration curve. In addition, the diffusion coefficients of the two dextran fractions in the mixture (2.53×10^{-10} and 1.73×10^{-11} m^2 s^{-1}) are equal within experimental errors to the values obtained for samples containing only one of the two fractions at the same total weight percent of dextran (2.54×10^{-10} and 1.65×10^{-11} m^2 s^{-1}, respectively). This indicates that microaveraging effects[23] are insignificant under our experimental conditions,[12] i.e., with the concentrations of the polysaccharidic solutions used in this study (see Experimental Section).

Such discrimination becomes obviously more difficult to achieve when the difference in the molecular weights of the two fractions reduces as illustrated with a mixture of two dextran fractions of M_W 270 and 670 kDa. Again, no microaveraging effects were observed. In the case of this mixture, a single exponential fit is not adequate to process the diffusion data and better results are obtained with a biexponential fit (Figure 4a). The biexponential fit of polydisperse polymers with the diffusion coefficient differing only by a factor of 2 is certainly not unique; in fact, a fitting procedure that takes account of polydispersity is likely to yield consistent results as well. However, the validity of the biexponential fit still holds in our case as we are dealing with only two monodisperse fractions (Figure 4b). In other words, the reliability of such a fitting procedure requires some a priori knowledge on the sample to analyze.[10]

Conclusion

DOSY has been proposed as a versatile tool for determining the average molecular weight of linear water soluble polysaccharides. The technique achieves results which show a good agreement with reported data obtained with other conventional methods. Moreover, DOSY allows for the determination of the molecular weight of the same samples used for the NMR structural analysis without any need of further manipulation, which could otherwise modify or even degrade the sample. In addition, when the experimental conditions are optimized, the measurement is rather fast. The good results achieved for uncharged water soluble polysaccharides encourage us to analyze with DOSY other types of polysaccharides, such as ionic polysaccharides, which require specific experimental conditions in order to be correctly analyzed.

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